

An Analysis of Differential privacy machine learning

Differential privacy machine learning

Differential privacy machine learning -- Part 1

Definition Let ϵ be a positive real number and A $\{\displaystyle \{\mathcal{A}\}\}$ be a randomized algorithm that takes a dataset as input (representing the actions of the trusted party holding the data). Let $\text{im } A$ $\{\displaystyle \{\text{trm } \{\mathcal{A}\}\}$ denote the image of A $\{\displaystyle \{\mathcal{A}\}\}$. The algorithm A $\{\displaystyle \{\mathcal{A}\}\}$ is said to provide (ϵ, δ) -differential privacy if, for all datasets D_1 $\{\displaystyle D_{\{1\}}\}$ and D_2 $\{\displaystyle D_{\{2\}}\}$ that differ on a single element (i.e., the data of one person), and all subsets S $\{\displaystyle S\}$ of $\text{im } A$ $\{\displaystyle \{\text{trm } \{\mathcal{A}\}\}$ $\{\mathcal{A}\}$:

Differential privacy machine learning -- Part 2

Toss a coin. If heads, then toss the coin again (ignoring the outcome), and answer the question honestly. If tails, then toss the coin again and answer "Yes" if heads, "No" if tails. (The seemingly redundant extra toss in the first case is needed in situations where just the act of tossing a coin may be observed by others, even if the actual result stays hidden.) The confidentiality then arises from the refutability of the individual responses. But, overall, these data with many responses are significant, since positive responses are given to a quarter by people who do not have the attribute A and three-quarters by people who actually possess it. Thus, if p is the true proportion of people with A , then we expect to obtain $(1/4)(1-p) + (3/4)p = (1/4) + p/2$ positive responses. Hence it is possible to estimate p . In particular, if the attribute A is synonymous with illegal behavior, then answering "Yes" is not incriminating, insofar as the person has a probability of a "Yes" response, whatever it may be. Although this example, inspired by randomized response, might be applicable to microdata (i.e., releasing datasets with each individual response), by definition differential privacy excludes microdata releases and is only applicable to queries (i.e., aggregating individual responses into one result) as this would violate the requirements, more specifically the plausible deniability that a subject participated or not.

Differential privacy machine learning -- Part 3

$\{(P_{FP}, P_{FN}) \mid P_{FP} + \epsilon P_{FN} \geq 1 - \delta, \epsilon P_{FP} + P_{FN} \geq 1 - \delta\}$ $\{\displaystyle \{(P_{\{\text{FP}\}}, P_{\{\text{FN}\}}) \mid P_{\{\text{FP}\}} + \epsilon P_{\{\text{FN}\}} \geq 1 - \delta, \epsilon P_{\{\text{FP}\}} + P_{\{\text{FN}\}} \geq 1 - \delta\}$

Differential privacy machine learning -- Part 4

ϵ -differentially private mechanisms Since differential privacy is a probabilistic concept, any differentially private mechanism is

necessarily randomized. Some of these, like the Laplace mechanism, described below, rely on adding controlled noise to the function that we want to compute. Others, like the exponential mechanism and posterior sampling sample from a problem-dependent family of distributions instead. An important definition with respect to ϵ -differentially private mechanisms is sensitivity. Let d $\{\displaystyle d\}$ be a positive integer, D $\{\displaystyle \{\mathcal{D}\}\}$ be a collection of datasets, and $f : D \rightarrow \mathbb{R}^d$ $\{\displaystyle \text{colon } \{\mathcal{D}\} \rightarrow \mathbb{R}^d\}$ be a function. One definition of the sensitivity of a function, denoted Δf $\{\displaystyle \Delta f\}$, can be defined by: $\Delta f = \max \|\mathbf{f}(D_1) - \mathbf{f}(D_2)\|_1$ $\{\displaystyle \Delta f = \max \|\mathbf{f}(D_{\{1\}}) - \mathbf{f}(D_{\{2\}})\|_1\}$, where the maximum is over all pairs of datasets D_1 $\{\displaystyle D_{\{1\}}\}$ and D_2 $\{\displaystyle D_{\{2\}}\}$ in D $\{\displaystyle \{\mathcal{D}\}\}$ differing in at most one element and $\|\cdot\|_1$ $\{\displaystyle \|\cdot\|_1\}$ denotes the L1 norm. In the example of the medical database below, if we consider f $\{\displaystyle f\}$ to be the function Q_i $\{\displaystyle Q_{\{i\}}\}$, then the sensitivity of the function is one, since changing any one of the entries in the database causes the output of the function to change by either zero or one. This can be generalized to other metric spaces (measures of distance), and must be to make certain differentially private algorithms work, including adding noise from the Gaussian distribution (which requires the L2 norm) instead of the Laplace distribution. There are techniques (which are described below) using which we can create a differentially private algorithm for functions, with parameters that vary depending on their sensitivity.

Differential privacy machine learning -- Part 5

Data utility and accuracy. The main concern with differential privacy is the trade-off between data utility and individual privacy. If the privacy loss parameter is set to favor utility, the privacy benefits are lowered (less "noise" is injected into the system); if the privacy loss parameter is set to favor heavy privacy, the accuracy and utility of the dataset are lowered (more "noise" is injected into the system). It is important for policymakers to consider the trade-offs posed by differential privacy in order to help set appropriate best practices and standards around the use of this privacy preserving practice, especially considering the diversity in organizational use cases. It is worth noting, though, that decreased accuracy and utility is a common issue among all statistical disclosure limitation methods and is not unique to differential privacy. What is unique, however, is how policymakers, researchers, and implementers can consider mitigating against the risks presented through this trade-off. Data privacy and security. Differential privacy provides a quantified measure of privacy loss and an upper bound

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and allows curators to choose the explicit trade-off between privacy and accuracy. It is robust to still unknown privacy attacks. However, it encourages greater data sharing, which if done poorly, increases privacy risk. Differential privacy implies that privacy is protected, but this depends very much on the privacy loss parameter chosen and may instead lead to a false sense of security. Finally, though it is robust against unforeseen future privacy attacks, a countermeasure may be devised that we cannot predict.

Differential privacy machine learning -- Part 6

Attacks in practice Because differential privacy techniques are implemented on real computers, they are vulnerable to various attacks not possible to compensate for solely in the mathematics of the techniques themselves. In addition to standard defects of software artifacts that can be identified using testing or fuzzing, implementations of differentially private mechanisms may suffer from the following vulnerabilities:

Differential privacy machine learning -- Part 7

21st century In 2003, Kobbi Nissim and Irit Dinur demonstrated that it is impossible to publish arbitrary queries on a private statistical database without revealing some amount of private information, and that the entire information content of the database can be revealed by publishing the results of a surprisingly small number of random queries—far fewer than was implied by previous work. The general phenomenon is known as the Fundamental Law of Information Recovery, and its key insight, namely that in the most general case, privacy cannot be protected without injecting some amount of noise, led to development of differential privacy. In 2006, Cynthia Dwork, Frank McSherry, Kobbi Nissim and Adam D. Smith published an article formalizing the amount of noise that needed to be added and proposing a generalized mechanism for doing so. This paper also created the first formal definition of differential privacy. Their work was a co-recipient of the 2016 TCC Test-of-Time Award and the 2017 Gödel Prize. Since then, subsequent research has shown that there are many ways to produce very accurate statistics from the database while still ensuring high levels of privacy.